

P1i: Theorems B and C of P1f at the small denominators $3 \leq N \leq 16$
of the arc $[0.143, 0.857]$, by per-case landscape certificates
research programme P1, ninth atom, 2026-09-24. Not yet reviewed.

Notation and numbering are those of P1f (Sections set, dec, est; Lemmas saddle, amp, classes, EM, head, LS, LSp; Theorems asym, B, C) and P1h (Theorems M', main', B', C'). Labels: PROVED; *computer-assisted* (reduced to finitely many Arb ball inequalities, checked by the named script, which fails loudly); VERIFIED (numerics, not a proof); HEURISTIC; OPEN. The level of rigour is that of P1f: the $O(t)$ of Laplace's method and the index j_0 are not made explicit.

Outcome. The family is every reduced a/N with $3 \leq N \leq 16$, $0.143 \leq a/N \leq 0.857$ and $a/N \neq \frac{1}{2}$. It has 56 members.

- (a) **Theorem 3 (PROVED, computer-assisted).** For each of the 52 members with $N \notin \{4, 6\}$, every conclusion of P1f Theorem B holds at $\zeta = e(a/N)$. So the zeros of B accumulate at ζ along the tie ray $\arg t = \theta^*$, they are simple, $t_j^* = t_j^0 + O(j^{-3})$, and they are simple poles of G_0^T . The analytic Lemmas LS, LSp, head of P1f are replaced by a per-case Arb certificate (P1i_landscape_cert.py: all 52 CERTIFIED, 34 s in total, 27 MB).
- (b) **Theorem 4 (PROVED, computer-assisted; standing thm:model, thm:RJ).** For the same 52 members, P1f Theorem C(i)–(iii) holds. So the poles of U and V accumulate at $\pm\sqrt{\zeta}, \pm\sqrt{\bar{\zeta}}$, along the square roots of the zeros of (a). The transfer conditions are re-certified for all 56 members (P1i_transfer.py; min certified modulus 0.2351).
- (c) $N = 4$ ($\frac{1}{4}, \frac{3}{4}$) is **prop:qi/cor:Ui** of paper2a, and is not redone here. The certificate reproduces the known obstruction there: $T = -0.0707 < 0$.
- (d) $N = 6$ ($\frac{1}{6}, \frac{5}{6}$): P1f Theorem B is *not* closed by this method. The tie height is $T = -0.10878 < 0$, below the head height 0 of $b_0 = 1$ (Section 5). The zeros exist numerically on the predicted tie ray (VERIFIED). A proof needs the $q = i$ technique of a contour through $\operatorname{Re} x < 0$ with theta reflection, generalised from 4 to 6 classes (OPEN).

Correction to the brief: what is new. The bare statement “the zeros of B accumulate at ζ ” is *already* a consequence of P1h. The accumulation set is closed, and P1h Theorem B' puts the zeros of B densely at every point of $\arg q \in [2\pi \cdot 0.143, 2\pi \cdot 0.857]$. Likewise, P1h Theorem C' (with closure) gives accumulation of the poles of U, V at $\pm\sqrt{\zeta}$ for every ζ with $a/N \in [\frac{1}{6}, \frac{5}{6}]$, and this includes $\frac{1}{6}$ and $\frac{5}{6}$. The new content is therefore the following:

- (i) the *direct* statement at each small- N point: an explicit sequence of simple zeros on the tie ray at ζ itself, with the two-saddle asymptotic, and the limit $t_1 \rightarrow t_1^* = -\frac{1}{2}(1 + \omega)$ of P1f Theorem C along it;
- (ii) accumulation of the poles of U, V at $\pm\sqrt{e(\pm\frac{2}{13})}$. Since $\frac{2}{13} = 0.1538 \in [0.143, \frac{1}{6})$, this is not implied by P1h. It is the only small- N member of $[0.143, \frac{1}{6}) \cup (\frac{5}{6}, 0.857]$.

1 Why the P1f constants fail at small N

In P1f the landscape inequalities are proved from crude, N -uniform bounds. Examples are $|R_\mu| \leq \frac{\pi^2}{6N^2} + \dots$, $\cos \theta^* \geq 0.9087$, $|w| \leq 2.3 \cdot 10^{-6}$, and the head constant 0.0837. All of these need $N \geq 16$ (P1h: $N\ell \gtrsim 9$). At $N = 3$, $\frac{\pi^2}{3N^2} = 0.366$ already exceeds $T = 0.3335$, $|w| = 0.024$, and $|\theta^*| = 22.9^\circ$. The *true* landscape is fine except at $N = 4, 6$ (VERIFIED first by the floating sampler `P1f_landscape_check.py`, then certified). So we keep the whole P1f architecture: the exact class decomposition (Lemma classes), the kernel expansion with $M = 2N$ modes, the EM lemma, Laplace, Rouché, and the transfer. Only Lemmas head, LS, LSp are replaced by the per-case Lemma 1 and Proposition 2. Lemma W of P1d needs $N\ell \geq 3$. Over the 52 members, $\min N\ell = 3.727$ (at $\frac{1}{3}$).

2 The head at fixed N

Lemma 1 (head $_N$; PROVED). *Let $N \geq 3$, $\ell > 0$, $\delta = \frac{1}{8N}$, $|\theta| < \frac{\pi}{2}$, $K = \lfloor \delta/r \rfloor$, and*

$$\varrho_N = 1 - \frac{e^{1/(4N)}}{4N \cdot 2 \sin(\pi/N)}, \quad H_N = \delta \left(\ell + 2 \log \frac{1}{\varrho_N} \right) + \frac{2\delta}{N} \left(1 + \log \frac{1}{1.8\delta} \right).$$

Then $\max_{k \leq K} r \log |b_k| \leq H_N + o(1)$ as $r \rightarrow 0$. So $|H| = |\sum_{k \leq K} b_k| \leq e^{(H_N + o(1))/r}$. The same holds for $\sum_{k \leq K} b_k q^{\pm k}$.

Proof. This is the proof of P1f Lemma head with the N -dependence kept.

- $|q^{k^2}| \leq 1$. Also $|c_t - c| = 2|1 - e^{-t}| \leq 2re^r$, so $k r \log |c_t| \leq \delta \ell + o(1)$.
- For $i \leq 2k \leq 2\delta/r$ with $N \nmid i$: $|1 - e^{-it}| \leq ir e^{ir} \leq \frac{e^{1/(4N)}}{4N}$ and $|1 - \zeta^i| \geq 2 \sin \frac{\pi}{N}$. Hence $|1 - \zeta^i e^{-it}| \geq \varrho_N |1 - \zeta^i|$, with $\varrho_N > 0$ for $N \geq 3$. There are at most $2k$ such factors, so they cost at most $2\delta \log(1/\varrho_N)$ in $r \log |b_k|$.
- The remaining product $\prod_{i \leq 2k, N \nmid i} |1 - \zeta^i|$ is $N^{M'} \geq 1$ over M' full periods, times a partial period $\geq P_N > 0$. This costs $r \log(1/P_N) = o(1)$.
- For $i = Nm$, $m \leq M' = \lfloor 2k/N \rfloor$: $|z| = Nmr \leq 2\delta = \frac{1}{4N} \leq \frac{1}{12}$, so $|1 - e^{-z}| \geq |z|(1 - \frac{|z|}{2} e^{|z|}) \geq 0.95Nmr$. With $\log M'! \geq M' \log M' - M'$ this costs $r \log \frac{1}{\prod_m 0.95Nmr} \leq f(rM')$, where $f(y) = y(1 + \log \frac{1}{0.95Ny})$. The function f is increasing for $y \leq \frac{1}{0.95N}$, and $rM' \leq \frac{2\delta}{N} = \frac{1}{4N^2}$, which gives the last term (with $1.9 \geq 1.8$).
- The factor $q^{\pm k}$ contributes $\leq kr^2 \leq \delta r = o(1)$.
- There are $K + 1$ terms, and $r \log(K + 1) = o(1)$.

□

3 The per-case landscape certificate

Fix a member a/N with $N \neq 4, 6$. Let $n, s, \theta^*, T, x_\mu, V_\mu, W_\mu$ be as in P1f (Section set), with n chosen by the P1f rule. Let $\Theta = [\theta^* - 10^{-5}, \theta^* + 10^{-5}]$ and $\mathcal{X}_A = \{\rho e^{i\theta} : \rho \in [\frac{\delta}{2}, 2\delta], \theta \in \Theta\}$; this contains every start point x_A^ρ of P1f Section dec for $r \leq \delta/(2N)$.

- The *explicit modes* of P1f Section dec:

- N odd: $\mu \in \{-(2N-1), \dots, 2N\}$, with remainders $\mu_- = -2N$, $\mu_+ = 2N+1$;
- N even, $\epsilon = +1$: the even $\mu \in [-(4N-2), 4N]$, with remainders $-4N$, $4N+2$;
- N even, $\epsilon = -1$: the odd $\mu \in [-(4N-1), 4N-1]$, with remainders $\mp(4N+1)$.

For $S_\pm$ with N even, ϵ flips (P1f Theorem C(i)).

- *Reference height*: $V_{\text{ref}} = V_n$ for B , and for $S_\pm$ with N odd. For $S_\pm$ with N even, $V_{\text{ref}} = V_{n+1}$ (height T'). Put $h_{\text{ref}}(\theta) = \text{Re}(V_{\text{ref}}e^{-i\theta})$.
- *Dominant modes*: $\{n, n+s\}$ for B and for $S_\pm$ with N odd, and $\{n+1\}$ for $S_\pm$ with N even.

Paths.

- Non-dominant explicit mode: $\Pi_\mu = [x_A, x_\mu] \cup (x_\mu + e^{i\theta/2}[0, \infty))$, as in P1f.
- Dominant mode: $\Pi_\mu = [x_A, P_\mu] \cup (x_\mu + e^{i\theta/2}[-\lambda_\mu, \infty))$ with $P_\mu = x_\mu - \lambda_\mu e^{i\theta/2}$ and $\lambda_\mu = \text{Re } x_\mu / (2 \cos \frac{\theta}{2})$. So the path crosses the saddle on its steepest-descent line. This change of path from P1f is needed at small N . At $\frac{1}{3}$ the straight segment $[x_A, x_\mu]$ meets the saddle at 80.5° from the steepest-descent direction, within 9.5° of a level direction. Ω_μ is not entire: it has a logarithmic/dilogarithm branch point from $\text{Li}_2(e^{-2Nx})$ at $e^{-2Nx} = 1$, i.e. on $\text{Re } x = 0$. Condition (d) of Proposition 2 certifies $\text{Re } x > 0$ on every box and ray of both the old and the new path, so Ω_μ is holomorphic on the simply-connected half-plane containing both paths, and Cauchy's theorem (not global entireness) gives the same integral for any path with the same endpoints there.
- Remainder modes: $\gamma_\mp = x_A + e^{i(\theta \mp \pi/8)}[0, \infty)$, as in P1f. The kernel bound is ≤ 2.09 by (iv) of `lem:qi-bounds`.

Proposition 2 (computer-assisted: `P1i_landscape_cert.py`). *Take $\theta \in \Theta$ and $x_A \in \mathcal{X}_A$ on the ray $\arg x_A = \theta$, and put $\eta = 10^{-3}$. For each of the 52 members and for B (and, for N even, $S_\pm$):*

(H) $H_N \leq h_{\text{ref}}(\theta) - \eta$.

(a) *For every non-dominant explicit μ : $\text{Re}((\Omega_\mu(x) - V_{\text{ref}})e^{-i\theta}) \leq -\eta$ on Π_μ .*

(b) *For every dominant μ :*

- $\text{Re } \Omega_\mu''(x_\mu + ve^{i\theta/2}) < 0$ for $|v| \leq \frac{1}{4}$;
- $\text{Re}((\Omega_\mu(x) - V_\mu)e^{-i\theta}) < 0$ on the rest of Π_μ , with a uniform negative bound on each piece outside $|v| < \frac{1}{4}$.

(c) *On $\gamma_\mp$: $\text{Re}((\Omega_{\mu_\mp}(x) - V_{\text{ref}})e^{-i\theta}) \leq -\eta$.*

(d) *$\text{Re } x > 0$ on every certified box. So $\text{Re } x \geq \delta' > 0$ on the compact pieces, and $\text{Re } x$ increases on the rays: $\cos \frac{\theta}{2} > 0$, and $\cos(\theta \mp \frac{\pi}{8}) > 0$ and $\cos(\theta \mp \frac{\pi}{4}) > 0$ are certified.*

Method (all in Arb, 128 bits). • w is the quadratic-formula root of $c^N w = (1-w)^2$. The script certifies $|w| < 1$ and the residual; $|w| \leq 0.024$ for all members.

- $x_\mu = \frac{L_\mu}{2} - \xi$ is an enclosure of the exact saddle (P1f Lemma saddle(i)). θ^* is enclosed, and the tie $h_n(\theta^*) = h_{n+s}(\theta^*)$ is checked.

- *Boxes*: on a box X , $\text{Re}((\Omega_\mu(X) - V)e^{-i\Theta})$ is bounded by the centred form $\text{Re}((\Omega_\mu(m) - V)e^{-i\Theta}) + \text{rad}(X) |\Omega'_\mu(X)|$, with m the exact midpoint. Li_2 is evaluated only at exact points. $\Omega'_\mu = L_\mu - 2x - \frac{2}{N} \text{Log}(1 - e^{-2Nx})$ and $\Omega''_\mu = -2 - 4/(e^{2Nx} - 1)$ are evaluated on balls.
 - Each piece is parametrised by one real interval. On the segments the point x_A is a ball, in four pieces of $[\frac{\delta}{2}, 2\delta]$.
 - Pieces are bisected adaptively until the bound is below the target. The target is $\min(-\eta, \frac{1}{2} \cdot \text{sampled maximum})$, or $\frac{1}{2} \cdot \text{sampled maximum}$ for dominant pieces, where the sampled maximum is itself an Arb upper bound at 49 points. Failure at depth 22 aborts the run.
- *Taylor (b)*: $\Omega'_\mu(x_\mu) = 0$ exactly, so for $x = x_\mu + ve^{i\theta/2}$

$$W_\mu(x) - h_\mu = \text{Re} \int_0^v (v - u) \Omega''_\mu(x_\mu + ue^{i\theta/2}) du \leq \frac{v^2}{2} \sup \text{Re} \Omega''_\mu < 0$$

(since $e^{-i\theta}(e^{i\theta/2})^2 = 1$). This is certified on sub-balls of $v \in [-\frac{1}{4}, \frac{1}{4}]$. The certified supremum is ≤ -0.96 over all members.

- *Ray tails, explicit modes* ($v \geq V_1$): P1f Lemma saddle(iii) gives $\Omega_\mu = V_\mu - (x - x_\mu)^2 - R_\mu$ with $|R_\mu| \leq C_0 + C_1v$, where $C_0 = (\frac{\pi^2}{6} + |\text{Li}_2(w)|)/N^2$ and $C_1 = 2|\text{Log}(1 - w)|/N$ (because $|\text{Li}_2(e^{-2Nx})| \leq \frac{\pi^2}{6}$ for $\text{Re } x \geq 0$). So the exponent is $\leq \text{Re}((V_\mu - V)e^{-i\theta}) + C_0 + C_1v - v^2$, which decreases for $v \geq C_1/2$. The script checks that this is $\leq$ the target at V_1 .
- *Ray tails, remainder modes* ($\varrho \geq P_1$): $\text{Re}(\Omega e^{-i\theta}) \leq \text{Re}((xL_\mu - x^2)e^{-i\theta}) + \frac{\pi^2}{3N^2}$. With $x = x_A + \varrho e^{i\beta}$, $\beta = \theta \mp \frac{\pi}{8}$, this is

$$A_0 + A_1\varrho - \cos(\theta \mp \frac{\pi}{4})\varrho^2, \quad A_1 = \text{Re}((L_\mu - 2x_A)e^{\mp i\pi/8}).$$

The script checks that the vertex lies below P_1 and that the value at P_1 is $\leq -\eta$.

The run prints, per member and per sum, upper bounds for the head margin $H_N - h_{\text{ref}}$, for $\max(a)$, for $\sup \text{Re} \Omega''$ (column `domQ`), for $\max(b)$ off the Taylor window (`domfar`), and for $\max(c)$. $\square$

Certified margins (P1i_certlog.txt; worst over the 52 members; all are Arb upper bounds):

- $H_N - T \leq -0.0789$ (at $\frac{2}{13}$);
- non-dominant ≤ -0.0166 (at $\frac{1}{5}$);
- dominant: $\sup \text{Re} \Omega'' \leq -0.96$ (at $\frac{2}{13}$), and off the window ≤ -0.0123 ($\frac{2}{13}$);
- remainder ≤ -0.0010 (at $\frac{1}{6}$ for $S_\pm$, not used; among the 52 members ≤ -0.0057 , at $\frac{1}{3}$).

The printed values are the loosest box bounds that met the target, not the true maxima. The floating sampler gives much larger true margins, e.g. -0.087 non-dominant at $\frac{1}{5}$. Fail-loud test: with $\eta = 0.1$ (P1I_ETA=0.1) the run at $\frac{2}{13}$ aborts with exit status 1.

Consistency with P1f: $\theta^* = 0^\circ$ at $\frac{3}{10}$ and 6.31° at $\frac{4}{11}$, as printed there. The rerun of $N = 16$ agrees with P1f/P1h, which already proved these members.

Table 1: Tie data for the certified members with $a/N < \frac{1}{2}$; conjugates have $\theta^* \mapsto -\theta^*$. From the certlog.

a/N	θ^*	T	T'	a/N	θ^*	T	T'
1/3	-22.85°	0.3335		3/11	-7.36°	0.3017	
1/5	20.18°	0.1636		4/11	6.31°	0.4128	
2/5	-13.23°	0.4264		5/11	-5.93°	0.4692	
2/7	11.13°	0.3148		5/12	10.96°	0.4091	0.4764
3/7	-9.36°	0.4527		2/13	-11.03°	0.0931	
3/8	16.72°	0.3229	0.4706	3/13	7.06°	0.2349	
2/9	-10.47°	0.2167		4/13	-5.79°	0.3519	
4/9	-7.26°	0.4636		5/13	5.23°	0.4320	
3/10	0°	0.2625	0.3612	6/13	-5.01°	0.4724	
2/11	10.49°	0.1446		3/14	-26.16°	0.1949	0.2401
				5/14	0°	0.3689	0.4193
				4/15, 7/15	$5.49^\circ, -4.34^\circ$	0.2944, 0.4744	
				3/16, 5/16, 7/16	$13.81^\circ, -9.28^\circ, 8.17^\circ$	0.1330, 0.3341, 0.4401	0.1704, 0.3722, 0.4788

4 The theorems

Theorem 3 (B at small N ; PROVED, computer-assisted). *Let $\zeta = e(a/N)$ with $3 \leq N \leq 16$, $N \notin \{4, 6\}$, $\gcd(a, N) = 1$, $0.143 \leq a/N \leq 0.857$, and $a/N \neq \frac{1}{2}$ (52 members). Then every conclusion of P1f Theorem asym and Theorem B holds at ζ .*

Proof. Run the proof of P1f Theorem asym with the following inputs:

- Lemma 1 and Proposition 2(H) in place of Lemma head. This gives $|H| \leq e^{(T(\theta) - \eta + o(1))/r}$.
- Proposition 2(a),(c),(d) in place of Lemma LS(a),(c),(d). With the EM lemma at $\delta' > 0$, this bounds all non-dominant explicit modes and both remainders by $C_N r^{-2} e^{(T(\theta) - \eta)/r}$.
- Proposition 2(b) in place of Lemma LS(b). On the dominant path the exponent is $\leq h_\mu - \kappa v^2$ near x_μ with $\kappa \geq 0.48$, and $< h_\mu$ uniformly away from it; the path crosses the saddle along its steepest-descent line. So Laplace's method [1] gives $\kappa A(\mu) e^{V_\mu/t} (1 + O(t))$ exactly as in P1f.

All inequalities hold on the box Θ , which gives the uniformity in $|\theta - \theta^*| \leq 10^{-5}$. Every other ingredient is independent of N or holds at fixed N : Lemmas classes and EM, the kernel expansion, Lemma amp (with $Z_N \neq 0$, Theorem M' of P1h, re-certified in `P1i_transfer.py`), and the Rouché step and Step 6 of `prop:minusone` in the proof of Theorem B. Lemma W holds since $N\ell \geq 3.72$. $\square$

Theorem 4 (C at small N ; PROVED, computer-assisted; standing `thm:model`, `thm:RJ`). *For the 52 members of Theorem 3, P1f Theorem C(i)–(iii) holds at the zeros q_j^* of Theorem 3. In particular $\pm\sqrt{q_j^*}$ and $\pm\sqrt{\bar{q}_j^*}$ are poles of U and of V for large j , and they accumulate at $\pm\sqrt{\zeta}$, $\pm\sqrt{\bar{\zeta}}$.*

Proof. • (i) For N odd, the landscape of $S_\pm$ is that of B (same modes, same classes, amplitude factor $e^{\mp x}$ subexponential), so Proposition 2 for B applies. The bracket equals 2 by the N -free Gauss-sum identity of P1f. For N even, Proposition 2 was run separately for $S_\pm$: the modes are $\mathcal{M}'$, the single dominant mode is $n + 1$, the reference height is T' , and ϵ is flipped. It replaces Lemma LSp.

- (ii) The identity $G^*(\mu + a) = e(-\mu/N)G^*(\mu - a)$ holds for all N .

- (iii) `P1i_transfer.py` (which calls `check` of `P1f_transfer_cert.py`) certifies $Z_N \neq 0$, $Z_N^+ \neq 0$ and the six ω -conditions for both roots at all 56 members, with min modulus 0.2351. At $\frac{1}{3}$ the factor with $1 - x + \zeta = 0$ reduces to $1/(2x) \neq 0$ (P1g), and the certificate encloses it directly. $\square$

Corollary 5. *The poles of U and V accumulate at $\pm\sqrt{e(\pm\frac{2}{13})}$. This point is not covered by P1h Theorem C' , which works on $[\frac{1}{6}, \frac{5}{6}]$ only.*

5 $N = 6$: exactly why it is not closed

At $\zeta = e(\frac{1}{6})$ we have $c = 2e^{2\pi i/3}$, $\ell = \log 2$, $\theta^* = 0$, tie pair $(-3, -1)$, and $T = -0.10878$ (certified enclosure; the head check fails with $H_6 - T \geq 0.1548$). In the decomposition $B = H + \sum(\text{integrals})$, the integrals are $e^{(T+o(1))/r} \rightarrow 0$ exponentially, while $H \ni b_0 = 1$. So $|H|$ is of order one unless it cancels. No bound of the form $|H| \leq e^{(T-\eta)/r}$ can hold term by term, whatever cutoff K is used; this is the same mechanism as at $q = i$, where $T = -0.0707$. The landscape itself is healthy: for $S_{\pm}$ ($T' = 0.1654$) the certificate passes at $\frac{1}{6}$ with every margin negative. So only the head blocks.

- **VERIFIED** (`P1f_zeros.py` 1 6, 30+ digits): zeros of B lie on the ray $\theta^* = 0$ at the two-saddle prediction, with $|1/t_j^* - u_j| |\Delta V| = 0.19, 0.032, 0.0082$ at $j = 4, 8, 12$. Theorem B's conclusion is numerically true at $\frac{1}{6}$.
- **OPEN**: a proof. The $q = i$ route (paper2a `prop:qi`: start at a negative half-integer s_c so that there is no head, and cross $\text{Re } x = 0$ with the theta reflection of `lem:qi-bounds(ii)`) would need that lemma for 6 residue classes and a new path certificate.

Accumulation of the zeros of B at $e(\frac{1}{6})$, and of the poles of U, V at $\pm\sqrt{e(\pm\frac{1}{6})}$, holds anyway by closure from P1h (see the correction above). What is open at $\frac{1}{6}$ is the direct tie-ray statement.

6 VERIFIED cross-checks

`P1f_zeros.py` (floating, 30+ digits) at $\frac{1}{3}$: $|1/t_j^* - u_j| |\Delta V| = 0.0109, 0.0052, 0.0025$ for $j = 4, 8, 16$, decaying like $|t_j|$, on $\theta^* = -22.85^\circ$; at $\frac{2}{13}$: $0.00084, 0.00042$ ($j = 4, 8$); at $\frac{1}{5}$: $0.0050, 0.0025, 0.0013$ ($j = 4, 8, 16$). The sampled floating check (P1f's sampler, run for all 56 members) agrees in sign with the certificate on every member, including the two failures at $N = 4, 6$.

7 What is not proved; weakest points

1. **Inherited level of rigour.** As in P1f and paper2a, the $O(t)$ of the EM lemma and of Laplace's method, and hence j_0 , are not explicit. The certificate proves the exponent inequalities with a uniform margin; that is exactly what the P1f argument uses.
2. **Transcription risk.** The script re-implements the mode bookkeeping of P1f Section dec: the explicit set, the remainder modes, and the flip of ϵ for $S_{\pm}$. It checks parity consistency and that the dominant modes are explicit, but the bookkeeping itself is taken from P1f and not re-derived here. It agrees with `P1f_landscape_check.py`.
3. **Path change.** The dominant paths differ from P1f (descent-line polygon). This is legitimate because Ω_{μ} is holomorphic on the half-plane $\text{Re } x > 0$ that condition (d) certifies both paths lie in, not because it is entire; the deformation is Cauchy's theorem on that half-plane. Reviewed by Reviewer AE (CORRECT, with this wording fixed).

4. **Range of x_A .** We cover $|x_A| \in [\delta/2, 2\delta]$ as P1f does. The true range is $\delta(1 + O(r))$.
5. **Thin printed margins.** These are artefacts of stopping the bisection early (e.g. remainder -0.0057 at $\frac{1}{3}$), not true tightness. Any positive margin suffices.
6. **Not closed:** $N = 6$ (Section 5). $N = 4$ is not re-proved here (`prop:qi`).
7. **Novelty is limited** (see the correction): the accumulation statements themselves, except U, V at $\frac{2}{13}$, already follow from P1h by closure.
8. **Standing:** python-flint/Arb (`polylog`, `exp`, `log` on balls), P1d/P1e/P1f/P1h as cited; for U, V also `thm:model`, `thm:RJ`.

Files (rootunity_zeros/general/P1i/): `P1i_landscape_cert.py`, `P1i_transfer.py`, `P1i_certlog.txt`. Every run used `perl -e 'alarm 290; exec @ARGV'`. The peak RSS was 27 MB (fresh pre-flight: 24 GB RAM, 5.4 GB disk free), and the files written total under 40 kB.

References

- [1] F. W. J. Olver, *Asymptotics and Special Functions*, Academic Press, 1974.